Additional Problems for Ch 3 HRW – Vectors
Addisionele Probleme vir Hfst 3 HRW – Vektore

	9^{th} Edition	8^{th} Edition
Questions	1, 5, 6	3, 7, 6
Problems	1, 2, 3, 5, 10, 25, 26, 33, 34, 52, 57, 63	3, 6, 1, 5, 14, 23, 30, 37, 36, 54, 71, 57.

1	3	(a) $a_x = -2.5$ m. (b) $a_y = -6.9$ m.	
2	6	(a) $r_x = 10.4$ m (b) $r_y = 6.0$ m	(a) $r_x = 13$ m. (b) $r_y = 7.5$ m.
3	1	(a) 47.2 m (b) 122°	
5	5	(a) 156 m (b) The answer can be phrased several equivalent ways: 129.8° counter-clockwise from east, or 39.8° west from north, or 50.2° north from west.	
10	14	(a) $r_x = 12$ m (b) $r_y = -5.8$ m (c) $r_z = -2.8$ m	
25	23	2.6 km.	
26	30	(a) $\vec{R} = (-3.18\text{ m})\hat{\text{i}} + (4.72\text{ m})\hat{\text{j}}$. (b) $\vert\vec{R}\vert = 5.69$ m. (c) $\theta = -56.0°$ (with $-x$ axis).	
33	37	(a) $\vert\vec{a}\times\vec{b}\vert = 12$ since the angle between them is 90º. (b) $\hat{\text{i}}\times\hat{\text{j}} = \hat{\text{k}}$, or the $+z$ direction. (c) $\vert\vec{a}\times\vec{c}\vert = 12$. (d) $-\hat{\text{i}}\times\hat{\text{j}} = -\hat{\text{k}}$, or the $-z$ direction. (e) $\vert\vec{b}\times\vec{c}\vert = 12$. (f) The vector points in the $+z$ direction	
34	36	(a) $\vec{a}\times\vec{b} = 2.0\hat{\text{k}}$. (b) $\vec{a}\cdot b = 26$ (c) $(\vec{a}+\vec{b})\cdot\vec{b} = 46$. (d) $a_b = \vec{a}\cdot\hat{b} = 5.8$.	
52	54	(a) $\vec{r} = \vec{d}_1 - \vec{d}_2 + \vec{d}_3 = (9.0\text{ m})\hat{\text{i}} + (6.0\text{ m})\hat{\text{j}} + (-7.0\text{ m})\hat{\text{k}}$. (b) $\vert\vec{r}\vert = 12.9$ m $\theta = 123°$. (c) $d_{\parallel} = -3.2$ m. (d) $d_{\perp} = 8.2$ m.	
57	71	$\vec{A} = 1.0\,\hat{\text{i}} + 4.0\,\hat{\text{j}}$. $A = \sqrt{(1.0)^2 + (4.0)^2} = 4.1$.	
63	57	(a) $\vec{d}_1\cdot(\vec{d}_2+\vec{d}_3) = 0 - 3.0 + 6.0 = 3.0\text{ m}^2$. (b) $\vec{d}_1\cdot(\vec{d}_2\times\vec{d}_3) = 52\text{ m}^3$. (c) $\vec{d}_1\times(\vec{d}_2+\vec{d}_3) = (11\hat{\text{i}} + 9.0\hat{\text{j}} + 3.0\hat{\text{k}})\text{ m}^2$	